

ABOYNE, United Kingdom

WMO: 030800

Lat: 57.0758N

Lon: 2.8411W

Elev: 459

StdP: 14.45

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 19.3	(c) 22.9	(d) 18.2	(e) 14.0	(f) 22.3	(g) 21.3	(h) 16.3	(i) 26.5	(j) 31.9	(k) 45.9	(l) 28.8	(m) 44.9	(n) 3.8	(o) 300	(p) 0.634

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 16.9	(c) 74.1	(d) 63.1	(e) 70.7	(f) 61.1	(g) 67.7	(h) 59.2	(i) 64.6	(j) 71.6	(k) 62.5	(l) 68.5	(m) 60.6	(n) 66.2	(o) 9.4	(p) 90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.0	84.6	66.9	59.8	78.1	65.2	58.0	73.0	63.5	29.9	71.6	28.4	68.5	27.0	66.3	83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 24.4	(b) 20.8	(c) 18.0	(e) 12.7	(f) 79.2	(g) 8.7	(h) 2.8	(i) 6.4	(j) 81.2	(k) 1.4	(l) 82.9	(m) -3.5	(n) 84.5	(o) -9.8	(p) 86.5		
(a) 24.4	(b) 20.8	(c) 18.0	(e) 13.7	(f) 68.6	(g) 4.6	(h) 4.6	(i) 10.3	(j) 71.9	(k) 7.6	(l) 74.6	(m) 5.0	(n) 77.2	(o) 1.7	(p) 80.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	46.1	37.0	37.9	40.3	43.9	48.4	53.3	57.3	56.3	52.7	47.0	41.2	37.3		
	DBStd	9.12	6.00	6.10	5.88	5.43	5.26	6.05	4.57	4.56	4.62	5.52	6.00	6.79		
	HDD50	2181	402	342	305	196	94	27	1	4	23	124	268	395		
	HDD65	6902	867	759	766	633	514	352	242	271	370	557	713	858		
	CDD50	762	1	3	4	12	45	125	228	200	104	32	6	2		
	CDD65	6	0	0	0	0	0	0	4	2	0	0	0	0		
	CDH74	106	0	0	0	0	5	16	44	39	2	0	0	0		
	CDH80	8	0	0	0	0	0	2	1	5	0	0	0	0		
(14) Wind	WSAvg	7.1	8.3	8.5	8.5	7.3	6.9	6.6	5.8	5.7	6.4	7.0	7.1	7.6		
(15) Precipitation	PrecAvg	36.0	3.9	2.3	2.2	2.6	2.6	2.6	3.0	3.0	2.8	3.9	3.9	3.3		
	PrecMax	46.5	10.4	4.6	4.0	6.8	5.3	5.6	5.6	6.5	10.7	8.5	8.5	6.7		
	PrecMin	24.1	0.6	0.6	0.4	0.4	0.7	0.9	1.3	0.8	0.9	0.7	1.9	1.5		
	PrecStd	4.9	2.0	1.2	0.8	1.7	1.1	1.4	1.2	1.4	2.0	1.9	2.0	1.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.0	58.1	61.5	66.7	72.8	76.0	78.6	79.5	72.5	64.5	59.0	56.6		
		MCWB	50.0	51.3	50.3	54.1	61.1	61.2	66.7	66.6	61.5	57.5	53.1	51.3		
	2%	DB	52.0	54.5	56.4	61.6	67.4	70.5	74.4	72.5	67.8	61.1	55.9	53.9		
		MCWB	47.4	48.5	48.6	51.8	56.8	60.8	63.7	62.9	59.3	54.9	51.2	49.4		
	5%	DB	49.4	51.3	53.3	57.7	63.2	66.9	70.7	68.7	64.6	58.6	53.3	51.1		
		MCWB	45.2	46.3	46.8	50.0	54.1	58.8	61.6	60.3	57.4	53.5	48.9	47.0		
	10%	DB	46.6	48.2	50.7	54.7	59.5	64.0	67.2	65.7	61.9	56.4	50.9	48.2		
		MCWB	42.9	43.6	45.1	48.0	51.8	56.7	59.5	58.3	55.6	51.9	47.4	44.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.4	52.4	51.5	56.0	62.7	65.7	69.4	68.8	63.3	58.4	54.4	51.9		
		MCDB	54.3	57.7	58.6	64.2	71.8	71.5	74.8	73.6	70.0	62.5	57.6	55.9		
	2%	WB	47.8	48.9	49.3	52.9	58.1	62.4	65.0	63.8	60.3	55.9	51.8	49.7		
		MCDB	51.7	53.9	55.5	59.7	65.8	67.8	71.8	70.7	66.0	59.8	55.4	53.7		
	5%	WB	45.5	46.6	47.4	50.6	55.0	59.7	62.8	61.6	58.3	54.3	49.5	47.3		
		MCDB	48.9	50.9	52.8	56.8	61.7	65.6	68.7	67.3	63.0	57.7	52.4	50.6		
	10%	WB	43.2	44.0	45.4	48.6	52.5	57.4	60.7	59.6	56.5	52.6	47.8	45.0		
		MCDB	46.3	47.8	50.4	54.1	58.4	63.1	65.9	64.4	60.8	55.7	50.4	47.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.0	12.7	14.6	16.8	18.5	16.8	16.9	17.3	16.6	14.3	12.4	11.5		
		MCDBR	13.7	15.9	19.8	22.8	25.7	23.2	23.3	22.2	21.4	17.2	14.1	14.7		
		MCWBR	11.4	12.1	13.3	14.5	16.2	14.8	14.1	13.8	14.0	12.8	11.5	12.4		
	5% WB	MCDBR	13.6	14.5	19.0	21.5	23.8	21.5	20.8	20.1	18.9	15.2	13.6	14.4		
		MCWBR	11.6	11.5	13.3	14.3	15.7	14.9	13.9	13.5	13.4	12.1	11.6	12.4		
(40) Clear-Sky Solar Irradiance	taub		0.300	0.323	0.351	0.373	0.372	0.370	0.372	0.361	0.345	0.331	0.303	0.285		
	taud		2.256	2.342	2.319	2.298	2.351	2.380	2.378	2.419	2.472	2.443	2.318	2.198		
	Ebn at Noon		181	231	254	266	274	276	272	268	256	226	180	149		
	Edh at Noon		16	23	30	35	36	35	35	31	25	20	15	13		
(44) All-Sky Solar Radiation	RadAvg		145	350	676	1105	1485	1461	1416	1118	785	417	189	103		
	RadStd		20	32	81	112	82	122	119	73	62	41	17	13		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (35 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page